

ABHAY SINGH SENGAR

AI / ML Engineer • Software Engineer • B.Tech CSE 2026

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PROFESSIONAL SUMMARY

Software engineer specializing in **production AI systems** — LLM applications, RAG pipelines, and multi-agent architectures — currently building MVP features as an AI / Full-Stack Engineering Intern at Brandora, with prior internship experience at WorkElate and a **Meta HackerCup 2025 AI Track Global Finalist** placement among the world's top AI engineers. Shipped end-to-end systems including AI agents for automated video and ad generation, a RAG-based compliance API (FastAPI, LangChain, GPT-4o), and a voice-powered AI interview platform with LangGraph multi-agent orchestration and IRT-adaptive scoring. B.Tech Computer Science Engineering from VIT Bhopal (CGPA 7.93), with strong fundamentals in **DSA, OOP, system design, and full-stack development** across Python, JavaScript, and SQL.

EDUCATION

VIT Bhopal University — Bhopal, India

Aug 2022 – Jul 2026

B.Tech in Computer Science Engineering • CGPA: 7.93 / 10.0 • Reg. 22BCE11620

Class of 2026

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Theory of Computation, Machine Learning, Artificial Intelligence, Cloud Computing.

TECHNICAL SKILLS

Programming Languages: Python, SQL

CS Fundamentals: Data Structures, Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

Frameworks & Libraries: FastAPI, LangChain, LangGraph, React, Node.js, Next.js, NumPy, Pandas, scikit-learn, PyTorch

AI / ML Engineering: LLM Engineering, RAG Pipelines, Multi-Agent Systems, Voice AI, NLP, Prompt Engineering, GPT-4o

Databases: PostgreSQL, MongoDB, MySQL, Supabase, Redis

Cloud & DevOps: GCP (Vertex AI), Docker, Git, GitHub Actions, REST APIs, Microservices, CI/CD

PROFESSIONAL EXPERIENCE

AI / Full-Stack Engineering Intern — Brandora

May 2026 – Present

- Building the company's MVP from the ground up as a core engineering hire, owning AI feature development and full-stack delivery across the product surface.
- Developing an **AI agent for automated video generation**, orchestrating LLM-driven scripting, scene composition, and rendering into an end-to-end pipeline that turns brand briefs into shippable video assets.
- Engineering a **product-based AI ad generation system** that ingests product context and generates targeted ad creatives, working across the prompt-engineering, backend orchestration, and frontend layers to deliver the feature end-to-end.

AI Intern — WorkElate

Feb 2026 – Mar 2026

- Designed and shipped a **RAG-based Compliance-Checking API** using FastAPI, LangChain (LCEL), and GPT-4o, automating regulatory clause verification across enterprise documents and reducing manual review time.
- Built an **email-intent classification microservice** (Python, FastAPI, LangChain) to triage inbound communications, applying prompt engineering and structured output parsing for production reliability.
- Engineered an **APScheduler-based daily digest pipeline** on MongoDB (boards/cards schema), delivering automated stakeholder reports and demonstrating end-to-end ownership from data layer to API to scheduled delivery.

KEY PROJECTS

Skill Assessment Agent — Voice AI Interview Platform

[\[Live Demo\]](#)

Python, FastAPI, LangGraph, GPT-4o, IRT, ESCO, React, TypeScript, PostgreSQL

- Architected a voice-powered AI interview system with a **LangGraph supervisor multi-agent architecture**, orchestrating specialised agents for question generation, scoring, and gap analysis.
- Implemented **Item Response Theory (IRT)-based adaptive questioning** that calibrates difficulty in real-time to candidate ability, and Bloom's Taxonomy 1–5 rubric scoring with confidence intervals for evidence-based evaluation.
- Integrated the **ESCO occupational taxonomy** for grounded skill-gap analysis, producing structured, audit-ready candidate reports — deployed end-to-end on Vercel.

SparsaOS — Multi-Agent CRM Pipeline

[\[Live\]](#)

Python, FastAPI, GPT-4o, asyncio, SSE, React 19, TypeScript, Supabase (PostgreSQL)

- Built a **3-agent GPT-4o CRM pipeline** with asyncio-based orchestration and Server-Sent Events (SSE) streaming, delivering live agent reasoning to the frontend with sub-second perceived latency.
- Engineered the full stack: React 19 + TypeScript frontend, FastAPI backend, and Supabase PostgreSQL persistence layer, including authentication, schema design, and deployment on Vercel.
- Demonstrated production engineering skills end-to-end — from system design and async concurrency patterns to cloud deployment and live monitoring.

Production Voice AI Agent — Telephony Customer Support

Python, FastAPI, WebSockets, Deepgram Nova-3, GPT-4o-mini, ElevenLabs, Twilio Media Streams

- Designed a real-time voice agent integrating **Deepgram Nova-3** (speech-to-text), GPT-4o-mini (reasoning), ElevenLabs (text-to-speech), and Twilio Media Streams via WebSockets for telephony-grade customer support.
- Iterated through two full architecture revisions to optimise end-to-end latency and barge-in handling — owning system design, debugging, and deployment for a production-ready agent.

ACHIEVEMENTS & RECOGNITION

- **Meta HackerCup 2025 — AI Track, Global Finalist:** ranked among the top competitive programming and AI engineering finalists worldwide in Meta's flagship contest.

CERTIFICATIONS & INDUSTRIAL TRAINING

- **AI & Edge Computing for Industry Applications** — L&T EduTech (Industrial Internship-Equivalent Program). Hands-on training in TinyML, TensorFlow Lite, and edge deployment of ML models on resource-constrained hardware for industrial use cases.
- **MongoDB Database Administrator** — FacePrep (Industrial Internship-Equivalent Program). End-to-end DBA curriculum covering schema design, indexing, replication, sharding, performance tuning, and backup/recovery on MongoDB.
- **AI / ML — Google Cloud Vertex AI** — Smart Bridge (Virtual GenAI Internship). Practical certification covering the ML lifecycle on GCP Vertex AI: data preprocessing, model training, evaluation, and managed cloud deployment.

ADDITIONAL INFORMATION

Languages: English (professional), Hindi (native). **Location Preference:** Open to relocation across India (Bangalore, Hyderabad, Pune, Chennai, Mumbai, NCR) and remote/international. **Availability:** Immediate joining post-graduation (July 2026).